

# claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

## Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a83c6f1d85d30ca41.jsonl`  
- SHA-256: `4dcd8c791fd133a9b9f50e68404df51d58970566b40e1991cfc9465160465ca9`  
- Source modified: `2026-06-16T03:20:17+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

## Transcript

### 1. user (2026-06-16T03:19:39.715Z)

#### Source Extractor

Research question: "Research question: \*\*What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?\*\* Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best \*recent\* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score,  $\text{mAP@tIoU}$ ). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

**\*\*URL:\*\*** <https://arxiv.org/abs/1911.00068>

**\*\*Title:\*\*** Confident Learning: Estimating Uncertainty in Dataset Labels (Northcutt et al., JAIR 2021 / Cleanlab)

**\*\*Found via:\*\*** Knowledge-distillation licensing boundary + label-noise from teacher search

## Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
  - be a concrete, checkable statement (not vague generalities)
  - include a direct quote from the source as support
  - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

**2. assistant (2026-06-16T03:19:41.807Z)**

I'll fetch the source page and extract the relevant claims.

**3. user (2026-06-16T03:19:55.354Z)**

## Confident Learning: Key Claims Summary

**\*\*Publication Date:\*\*** Journal of Artificial Intelligence Research (JAIR) 2021; arXiv submitted October 31, 2019

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## (1) What is Confident Learning?

CL is a framework that "focuses instead on label quality by characterizing and identifying label errors in datasets, based on the principles of pruning noisy data, counting with probabilistic thresholds to estimate noise, and ranking examples to train with confidence."

The method operates on three core principles: removing suspect examples, using probabilistic counting to quantify label noise, and ordering samples by confidence for training prioritization.

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## (2) Theoretical Guarantees

The paper presents a "generalized CL which is provably consistent" and establishes "sufficient conditions where CL exactly finds label errors." Specifically, it works by constructing "the joint distribution between noisy (given) labels and uncorrupted (unknown) labels" under a class-conditional noise process assumption.

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## (3) Empirical Results

Key findings include:

- **CIFAR:** "CL performance exceeding seven recent competitive approaches for learning with noisy labels"
- **ImageNet:** Identified ontological overlap, estimating "645 'missile' images are mislabeled as their parent class 'projectile'" and achieved "moderately increase model accuracy (e.g., for ResNet) by cleaning data prior to training"
- **MNIST & Amazon Reviews:** Found errors in MNIST and improved sentiment classification

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## (4) Model-Agnostic Design

Yes. The framework is "not coupled to a specific data modality or model," requiring only predicted probability estimates rather than specific architectures.

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## (5) Software

"Open-source cleanlab release" is available for reproducible results.

### 4. assistant (2026-06-16T03:19:59.388Z)

The fetch succeeded. This is the primary research paper (Northcutt, Jiang, Chuang ? JAIR 2021), the foundational Confident Learning / Cleanlab paper. Let me assess and extract claims relevant to the research question, specifically Dimension 3 (label-noise handling from a teacher/pseudo-labels) and Dimension 4 (label-noise pitfalls at small n).

### 5. user (2026-06-16T03:20:09.297Z)

Structured output provided successfully

### 6. assistant (2026-06-16T03:20:17.349Z)

Extracted and submitted. Summary: this is the primary, peer-reviewed Confident Learning paper (Northcutt et al., JAIR 2021), the canonical citation behind the open-source `cleanlab` library. It is highly relevant to the research question's Dimension 3 (label-noise handling when

exploiting a teacher like Gemini for weak/pseudo-labels) and Dimension 4 (label-noise pitfalls at small  $n$ ). Key takeaways for the constrained setting: CL is model- and modality-agnostic, needing only out-of-sample predicted probabilities ? so it can audit the tiny golden set or any Gemini-derived pseudo-labels without coupling to the scorer; it has provable consistency guarantees under class-conditional noise; and it empirically beat seven noisy-label baselines on CIFAR with accuracy gains from pre-training data cleaning. One license caveat worth flagging downstream: cleanlab itself is open source, but its license should be verified against the MIT/Apache/BSD constraint before adoption (the fetch did not state the specific license).